

1. Set-up: HINs, network schemas and meta-paths

Heterogeneous information network (HIN). Following Sun et al. (*PathSim*, PVLDB 4(11):992–1003, 2011) and the survey of Shi, Li, Zhang, Sun & Yu (*A Survey of Heterogeneous Information Network Analysis*, IEEE TKDE 29(1):17–37, 2017), a **heterogeneous information network** is a graph

$$G = (V, E, \phi, \psi), \quad \phi : V \rightarrow \mathcal{A}, \quad \psi : E \rightarrow \mathcal{R},$$

in which every node $v \in V$ carries exactly one *node type* $\phi(v)$ from the type set $\mathcal{A}$, every edge $e \in E$ carries exactly one *relation type* $\psi(e)$ from $\mathcal{R}$, and

$$|\mathcal{A}| + |\mathcal{R}| > 2$$

This inequality *is* the definition of heterogeneity: when $|\mathcal{A}| = |\mathcal{R}| = 1$ the network degenerates to the homogeneous graph of Problem 1. Each relation is typed at both ends, inducing a map $A_i \xrightarrow{R} A_j$ between two specific node types; its inverse R^{-1} is in general a *different* relation (*writes* vs. *written-by*), which is why the released benchmarks below count 6 or 8 “edge types” where a reader might expect 3 or 4.

Network schema. The **network schema** $T_G = (\mathcal{A}, \mathcal{R})$ is the directed graph whose vertices are the node types $\mathcal{A}$ and whose edges are the relation types $\mathcal{R}$. It is a meta-template of which G is an instantiation: every node and edge of G maps to some type in T_G , so the schema states which kinds of connection the dataset can express at all. This is the object Section ?? draws for each benchmark. Note that the drawing carries real information here whereas it does not in the homogeneous case: Cora’s schema is a single node type with one self-relation, while DBLP’s is a four-type star.

Meta-path. A **meta-path** $\mathcal{P}$ of length ℓ is a path defined *on the schema*,

$$\mathcal{P} : A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \cdots \xrightarrow{R_\ell} A_{\ell+1},$$

denoting the composite relation $R = R_1 \circ R_2 \circ \cdots \circ R_\ell$ from type A_1 to type $A_{\ell+1}$. A node sequence $v_1 v_2 \cdots v_{\ell+1}$ in G with $\phi(v_i) = A_i$ and $\psi(v_i v_{i+1}) = R_i$ is a *path instance*. Meta-paths are written by type initials, so on DBLP APA abbreviates Author $\xrightarrow{\text{writes}}$ Paper $\xrightarrow{\text{written-by}}$ Author.

Meta-paths matter because each defines a *different* homogeneous projection of the same HIN, and the projections carry different semantics: on DBLP APA means *co-authorship* while APCPA means *published at the same venue*, and two authors may be strongly related under one and unrelated under the other. Nearly every method in Section ?? is a different answer to “how should meta-path structure be used?” — fixed and hand-chosen (metapath2vec, HAN), aggregated with attention (MAGNN), learned (GTN), or discarded in favour of per-relation parameters (R-GCN, HGT, simple-HGN).